REGULARITIES FOR A MINIMIZER OF SOME FUNCTIONALS WITH A CONVEXITY CONSTRAINT

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Abstract. This note presents a simplified version of the recent article authored by Wang and Zhou (*arXiv:2403.04235*), with a focus on elucidating the main idea behind their method. Within this concise exposition, we establish the $C^{1,\alpha}_{loc}$ regularity for minimizers of certain functionals subject to convexity constraints.

1. Introduction

In this short note, we will investigate the regularities for the minimizer of the functional

$$\int_\Omega F(x, u, Du)\ \mathrm{d}x, \tag{1.1}$$

over the set of convex functions, where Ω is a bounded, smooth, and convex domain in $\mathbb{R}^n$, $F(x, z, \mathbf{p}) : \mathbb{R}^n \times \mathbb{R} \times \mathbb{R}^n \to \mathbb{R}$ is a smooth function that is convex in each of the variables $z \in \mathbb{R}$ and $\mathbf{p} = (p_1, \ldots, p_n) \in \mathbb{R}^n$. Unlike the unconstrained case, the regularity of (1.1) is more complicated since the usual techniques in calculus of variations are no longer appropriate. In fact, because of the convexity constraint, it is generally difficult to write down a tractable Euler-Lagrange equation for the minimizers of (1.1) [Ca, CL2, Li]. There are some works related to finding suitably explicit approximations of the minimizers for practical purposes [CR, Le1, LZ, Le2], but almost no regularities of the minimizers are given in these papers. Thus, it may be impossible to obtain the regularity of the minimizers of (1.1) under convexity constraints for general $F(x, z, \mathbf{p})$, which inspires us to make some structural assumptions about $F(x, z, \mathbf{p})$ so that there are some other techniques that could be used to obtain regularities.

A typical form of $F(x, z, \mathbf{p})$ is from investigating the Rochet-Choné model [RC] in the monopolist's problem in Economics. When the cost functions is a quadratic function $|\mathbf{p}|^2/2$ in the Rochet-Choné model, we know that $F(x, z, \mathbf{p}) = (|\mathbf{p}|^2/2 - x \cdot \mathbf{p} + z)f(x)$. Then the C^1 regularity of the minimizer was proved by Carlier and Lachand-Robert [CL1]. Later, the interior $C^{1,1}$ regularity result was derived by Caffarelli and Lions through a very elegant argument in an unpublished notes [CL] (see [MRZ, Theorem 6] for a restatement). For more general form of $F(x, z, \mathbf{p})$, it can be from principal-agent problems. Principal-agent problems are a class of economic models with applications

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to tax policy, regulation of public utilities, product line design and contract theory, see details in [FKM, MRZ]. Then, under a certain uniformly "convex" of the cost function, the C^1 result in this case was extended by Chen [Ch1, Ch2], and the interior $C^{1,1}$ regularity was extended recently by McCann, Rankin and Zhang [MRZ]. The main technique in both results is still from Caffarelli and Lions [CL].

It's worth mentioning that in all the results derived previously, they assumed some uniformly "convex" properties of the cost function. While this assumption is reasonable for analyzing regularities, similar to the necessity of uniform ellipticity in studying elliptic equations, there still exist a few scenarios where certain degenerate elliptic equations exhibit limited regularities. Therefore, it might be beneficial to consider a slightly more general form of $F(x, z, \mathbf{p})$. One motivation for this consideration arises from the case of q-power cost ($1 < q < \infty$) in the Rochet-Chon'e model, where we have:

$$F(x, z, \mathbf{p}) = (|\mathbf{p}|^q/q - x \cdot \mathbf{p} + z)\eta_0(x),$$

where η_0 is the nonnegative relative frequency of agents in the population, see [RC, P790].

In this paper, we investigate the interior $C^{1,\alpha}$ regularity for the minimizer of functional

$$L[u] := \int_\Omega \left[F^1(x, Du(x)) + F^0(x, u(x))\right] \mathrm{d}x \tag{1.2}$$

with the admissible set

$$U_0 := \{u : \Omega \to \mathbb{R} \,|\, u \text{ is convex, } u(x) \geq a_0 + b_0 x\},$$

for some constants a_0, b_0.

To articulate our regularity result, we must impose some conditions on $F^1(x, \mathbf{p})$ and $F^0(x, z)$. Initially, a straightforward approach is to assume $F^1(x, \mathbf{p}) \sim C|\mathbf{p}|^q$ and $F^0(x, z) \sim Cz$, but such assumptions might be overly simplistic. To achieve a broader scope, we draw inspiration from Chen's [Ch1, Condition 1] and Le's [Le2, (1.5)&(1.6)]. Hence, to establish more general results, we introduce the following assumptions:

(H1) Fixed $q > 1$, $\exists\, \delta > 0$, such that $F^1(x, \mathbf{p}) - \delta|\mathbf{p}|^q$ is convex respect to $\mathbf{p}$;
(H2) There exists $M > 0$, such that $|D_z F^0(x, z)| \leq M$ for all $x \in X$ and $z \in \mathbb{R}$;
(H3) There exists $C_0 > 0$, such that

$$\left|D_{\mathbf{p}} F^1(x, \mathbf{p})\right| \leq C_0(|\mathbf{p}|^{q-1} + 1), \quad \left|D_{x_i p_i} F^1(x, \mathbf{p})\right| \leq C_0(|\mathbf{p}|^{q-1} + 1)$$

for all $x \in \Omega$, $\mathbf{p} \in \mathbb{R}^n$, and for each i.

Now, we state our main theorem as follows:

Theorem 1.1. *Assume that $\Omega \subset \mathbb{R}^n$ is a bounded convex domain. Suppose $F^1(x, \mathbf{p})$ and $F^0(x, z)$ satisfy* **(H1)**-**(H3)**. *Let u be a minimizer of the functional* (1.2). *Then $u \in C^{1,\alpha}_{loc}(\Omega)$, where $\alpha = \frac{1}{q-1}$ for $q > 2$ and $\alpha = 1$ for $1 < q \leq 2$.*

Remark 1.2. *When $q = 2$, we partially recover the results derived in* [CL, MRZ]. *When $q \geq 2$, examples can show that the regularity is optimal (Section 3).*

The basic idea for the proof of Theorem 1.1 is a perturbation argument used in the unpublished notes [CL], and there are some extensions in the recent paper [MRZ]. In [MRZ], the authors simplified Caffarelli and Lions's argument and extended it to the principal-agent problem by combining some results and methods from the optimal transport literature. Therefore, we will follow a similar framework as in the proof of [MRZ] to give the proof of Theorem 1.1, using a key lemma(Lemma 2.1). Our main new idea in this paper is that we are more meticulous in the choice of the comparison function and that we include a more delicate analysis. Specifically, in [MRZ], the authors perturb a support function of the minimizer to make the geometry of the section controllable, but the magnitude of the perturbation is not a concern for them. However, in our proof we need to introduce a small perturbation to the minimizer's support function in order to refine the section geometry. The small size of the perturbation is a crucial step in our proof (see the proof of Lemma 2.1 for details). This comparison function allows us to handle more general functionals that may be degenerate. This is one of the reasons why we can make assumptions **(H1)**-**(H3)** that cover all q-power cost ($1 < q < \infty$) functions in the Rochet-Choné models, while the assumption of certain types of uniform "convexity" is required in both [CL] and [MRZ].

We'll then provide a straightforward one-dimensional example to demonstrate that our regularities for the minimizer are optimal for $q > 2$. This issue has already been tackled by [MRZ, Remark 5] for $q = 2$. Regrettably, we couldn't locate an example indicating the optimality of regularity for the minimizer when $1 < q < 2$. This aspect remains open for investigation in the future.

The rest of the paper is organized as follows. Firstly, in Section 2.1, we establish Theorem 1.1 assuming the validity of Lemma 2.1. Subsequently, the proof of Lemma 2.1 is presented in Section 2.2. Lastly, in Section 3, an example is provided to showcase the optimal regularities of the minimizer of (1.2).

2. The proof of Theorem 1.1

2.1. **Proof of Theorem 1.1 provided a key lemma.** We will prove Theorem 1.1 by following the same method used in [MRZ] and the unpublished notes [CL]. Thus, the following lemma is very crucial.

Lemma 2.1. *Let $\Omega' \subset\subset \Omega$ and $d = \operatorname{dist}(\Omega', \partial\Omega)$. Then there exist $r_0 > 0$ and constants C_1, $C_2 > 0$ depending only on d, q, δ, C_0 and M such that the following property holds: If $u : \Omega \to \mathbb{R}$ is convex and $x_0 \in \Omega'$, then for any $r < r_0$ and*

$$h := \sup_{B_r(x_0)} [u(x) - (Du(x_0) \cdot (x - x_0) + u(x_0))]$$

is positive, there is an affine function $p_y(x) = x \cdot y + a$ such that

(1) The section $S := \{x \in \Omega \,|\, u(x) < p_y(x)\}$ has positive measure.
(2) On S, we have

$$\sup_{x \in S}(p_y(x) - u(x)) \leq h. \tag{2.1}$$

(3) There holds

$$\frac{1}{|S|}\int_S \big(F^1(x,y)-F^1(x,Du(x))\big)\ \mathrm{d}x \le C_1 h - C_2\frac{h^q}{r^q}. \tag{2.2}$$

Once we establish Lemma 2.1, Theorem 1.1 is proved immediately following a short argument.

Proof of Theorem 1.1. Fix $\Omega' \subset\subset \Omega$, $x_0 \in \Omega'$. Let $p_0(x) = Du(x_0)\cdot(x-x_0)+u(x_0)$ be the support function of u at x_0. By Lemma 2.2, it suffices to show that for any r less than a given r_0 (independent of u) there is

$$\sup_{B_r(x_0)} |u-p_0| \le Cr^{1+\alpha}$$

for some $\alpha > 0$ depends on q. Let

$$h = \sup_{B_r(x_0)} (u-p_0).$$

If $h=0$, then we are done. Thus, we assume $h>0$. Then by Lemma 2.1 there exists an affine function p_y and the associated section S. We set

$$u_h := \max\{u, p_y\}.$$

It is clear that u_h is convex, then there is

$$L[u_h] \ge L[u]$$

since u is a minimizer of (1.2). Note that u_h is different from u only on S, we have by **(H2)**, (2.1), (2.2) that

$$\begin{aligned}
0 &\le L[u_h]-L[u]\\
&= \int_S \big[(F^1(x,y)+F^0(x,p_y)) - (F^1(x,Du)+F^0(x,u))\big]\ \mathrm{d}x\\
&= \int_S (F^1(x,y)-F^1(x,Du))\ \mathrm{d}x + \int_S (F^0(x,p_y)-F^0(x,u))\ \mathrm{d}x\\
&\le \left(C_1 h - C_2\frac{h^q}{r^q} + Mh\right)|S|,
\end{aligned}$$

which gives us that

$$h \le Cr^{\frac{q}{q-1}},$$

i.e.

$$\sup_{B_r(x_0)} |u-p_0| \le Cr^{1+\frac{1}{q-1}}.$$

Then the proof is completed. □

In the above proof, we used a criterion for $C^{1,\alpha}$ regularity lemma for convex functions, which says that if a convex function separates its supporting planes in a $C^{1,\alpha}$ fashion, then it is indeed of class $C^{1,\alpha}$. This lemma can be found in many references, see, for

example, in Figalli's book [Fi, Lemma A.32]. For readers' convenience, we include the whole lemma.

Lemma 2.2 ([Fi, Lemma A.32])**.** *Let Z be an open convex set satisfying*

$$B_r(\bar{x}) \subset Z \subset B_R(\bar{x})$$

for some $0 < R \leq R$ and $\bar{x} \in \mathbb{R}^n$. Let $u : Z \to \mathbb{R}$ be a convex function, and assume that there exist constants K, C, $\bar{\varrho} > 0$ and $\alpha \in (0, 1]$ such that the following holds: u is $K-$Lipschitz in Z, and for every $x \in Z$ there exists $p_x \in \partial u(x)$ satisfying

$$u(z) - u(x) - p_x \cdot (z - x) \leq C|z - x|^{1+\alpha}, \quad \forall\, z \in Z \cap B_{\bar{\varrho}}(x).$$

Then $u \in C^{1,\alpha}$ with

$$\|Du\|_{C^\alpha(Z)} \leq \bar{C} = \bar{C}\Big(r, R, K, C, \bar{\varrho}\Big).$$

2.2. **The proof of Lemma 2.1.** Now, the remainder thing is to prove Lemma 2.1. The key idea comes from Caffarelli and Lions [CL], a similar argument can also be seen in [Ch1, Ch2, MRZ].

Proof of Lemma 2.1. After applying some transformations, we may assume that

$$x_0 = 0, \quad u(0) = 0, \quad \text{and} \quad Du(0) = 0.$$

Next, without loss of generality, we assume that

$$h = \sup_{B_r} u$$

occurs at re_1, i.e. $u(re_1) = h$. Because u attains a maximum over ∂B_r at re_1, it has 0 tangential derivative. Then we have $Du(re_1) = \kappa e_1$ for some $\kappa \geq h/r$ (by the convexity of u). Denote $y_\varepsilon := \frac{\varepsilon h}{r} e_1$ for some $\varepsilon \in (0, 1)$ to be determined later. Then the desired affine function will be

$$p_{y_\varepsilon}(x) = (x - re_1) \cdot y_\varepsilon + u(re_1).$$

Since

$$p_{y_\varepsilon}(0) - u(0) = -re_1 \cdot \frac{\varepsilon h}{r} e_1 + h = (1 - \varepsilon)h > 0,$$

which implies that the section $S := \{x \in \Omega \,|\, u(x) < p_{y_\varepsilon}(x)\}$ has positive measure.

Note that by Loper's maximum principle [Lo, Theorem 3.2], there is

$$(x - re_1) \cdot y_\varepsilon \leq \max\{0, (x - re_1) \cdot Du(re_1)\} \leq u(x), \quad x \in S.$$

Hence

$$p_{y_\varepsilon}(x) - u(x) = (x - re_1) \cdot y_\varepsilon + h - u(x) \leq h, \quad x \in S,$$

which yields (2.1).

Next, we show that

$$S \subset \{x \in \mathbb{R}^n \,|\, (1 - \varepsilon^{-1})r \leq x_1 \leq r\}. \tag{2.3}$$

Indeed, for $x = (r, x')$ there is

$$p_{y_\varepsilon}(x) - u(x) = (x - re_1) \cdot \frac{\varepsilon h}{r} e_1 + u(re_1) - u(x) = u(re_1) - u(x)$$

$$= (x - re_1) \cdot Du(re_1) + u(re_1) - u(x) \leq 0.$$

Hence we know that $x = (r, x') \notin S$. Similarly, for $x = ((1-\varepsilon^{-1})r, x')$ there is

$$\begin{aligned} p_{y_\varepsilon}(x) - u(x) &= (x - re_1) \cdot \frac{\varepsilon h}{r} e_1 + u(re_1) - u(x) \\ &= -\varepsilon^{-1} re_1 \cdot \frac{\varepsilon h}{r} e_1 + u(re_1) - u(x) \\ &= -u(x) \leq 0, \end{aligned}$$

which implies $x = ((1-\varepsilon^{-1})r, x') \notin S$. Thus, (2.3) follows by the convexity of S.

Now, we prove (2.2). Let $G(x, \mathbf{p}) = F^1(x, \mathbf{p}) - \delta|\mathbf{p}|^q$, then we have

$$\begin{aligned} \int_S (F^1(x, Dp_{y_\varepsilon}) - F^1(x, Du))\, \mathrm{d}x = \delta \int_S (|Dp_{y_\varepsilon}|^q - |Du|^q)\, \mathrm{d}x \\ + \int_S (G(x, Dp_{y_\varepsilon}) - G(x, Du))\, \mathrm{d}x. \end{aligned} \tag{2.4}$$

By (**H1**), we know that $G(x, \mathbf{p})$ is convex respect to $\mathbf{p}$. Hence, we have $G(x, Du) \geq G(x, Dp_{y_\varepsilon}) + D_{\mathbf{p}} G(x, Dp_{y_\varepsilon}) \cdot (Du - Dp_{y_\varepsilon})$. Note that $|Du|^q \geq \frac{1}{2^{q-1}}|Du - Dp_{y_\varepsilon}|^q - |Dp_{y_\varepsilon}|^q$, combining this and (2.4) we have that

$$\begin{aligned} \int_S (F^1(x, Dp_{y_\varepsilon}) - F^1(x, Du))\, \mathrm{d}x \leq \delta \int_S \left(2|Dp_{y_\varepsilon}|^q - \frac{1}{2^{q-1}}|Du - Dp_{y_\varepsilon}|^q \right) \mathrm{d}x \\ + \int_S D_{\mathbf{p}} G(x, Dp_{y_\varepsilon}) \cdot (Dp_{y_\varepsilon} - Du)\, \mathrm{d}x \end{aligned}$$

Hence, to prove (2.2), we need to estimate the terms

$$\int_S |Du - Dp_{y_\varepsilon}|^q\, \mathrm{d}x, \tag{2.5}$$

and

$$\int_S D_{\mathbf{p}} G(x, Dp_{y_\varepsilon}) \cdot (Dp_{y_\varepsilon} - Du)\, \mathrm{d}x. \tag{2.6}$$

We estimate (2.6) first. By the divergence theorem, we have

$$\begin{aligned} \int_S D_{\mathbf{p}} G(x, Dp_{y_\varepsilon}) \cdot (Dp_{y_\varepsilon} - Du)\, \mathrm{d}x &= \int_S D_{\mathbf{p}} G(x, Dp_{y_\varepsilon}) \cdot D_x(p_{y_\varepsilon} - u)\, \mathrm{d}x \\ &= \int_{\partial S \cap \partial\Omega} (p_{y_\varepsilon} - u) D_{\mathbf{p}} G(x, y_\varepsilon) \cdot \mathbf{n}\, \mathrm{d}S - \int_S (p_{y_\varepsilon} - u) \mathrm{div}_x (D_{\mathbf{p}} G(x, y_\varepsilon))\, \mathrm{d}x. \end{aligned} \tag{2.7}$$

By (**H3**), we know

$$\begin{aligned} |D_{\mathbf{p}} G(x, y_\varepsilon)| &\leq \left| D_{\mathbf{p}} F^1(x, y_\varepsilon) \right| + \delta q |y_\varepsilon|^{q-1} \\ &\leq (C_0 + \delta q)|y_\varepsilon|^{q-1} + C_0, \end{aligned}$$

and

$$|\mathrm{div}_x(D_{\mathbf{p}}G(x,y_\varepsilon))| = |\mathrm{div}_x(D_{\mathbf{p}}F^1(x,y_\varepsilon))| \le C_0|y_\varepsilon|^{q-1} + C_0.$$

By $y_\varepsilon = \frac{\varepsilon h}{r}$, (2.1), and (2.7), we have

$$\int_S D_{\mathbf{p}}G(x, Dp_{y_\varepsilon}) \cdot (Dp_{y_\varepsilon} - Du)\ \mathrm{d}x \le Ch|S| + C\frac{h^q}{r^{q-1}}|S|, \tag{2.8}$$

where we should use the estimate $|\partial S \cap \partial\Omega| \le C|S|$, which was proved by Carlier and Lachand-Robert [CL1] and Chen [Ch1, P82].

Next, we estimate (2.5). For $x = (x_1, x')$, we let $P(x) := (0, x')$ be the projection onto $\{x \in \mathbb{R}^n \,|\, x_1 = 0\}$. Denote $\frac{1}{K}S$ as the dilation of S by a factor $\frac{1}{K}$ with respect to the origin. Choosing $K = 2\,\mathrm{diam}(\Omega)/d$, where $d = \mathrm{dist}(\Omega', \partial\Omega)$. Hence $P(\frac{1}{K}S) + \frac{d}{2}e_1 \subset \mathrm{int}\,\Omega$. Taking r_0 smaller depending on d to ensure $S \subset \{x : x_1 \le \frac{d}{2}\}$. For $(0, x') \in P(\frac{1}{K}S)$, we let $l_{x'}$ be the line segment with greater x_1 component of the set $(P^{-1}(\frac{1}{K}S) \cap S)\backslash(\frac{1}{K}S)$ and write $l_{x'} = [a_{x'}, b_{x'}] \times \{x'\}$ where $b_{x'} > a_{x'}$. Then we have that the point $(b_{x'}, x')$ satisfies $b_{x'} \le \frac{d}{2}$, hence $(b_{x'}, x') \in \partial S \cap \mathrm{int}\,\Omega$.

Note that $u(0) - p_{y_\varepsilon}(0) = -(1-\varepsilon)h$ and $u(Ka_{x'}, Kx') - p_{y_\varepsilon}(Ka_{x'}, Kx') \le 0$, then by the convexity of $u - p_{y_\varepsilon}$, we have

$$u(a_{x'}, x') - p_{y_\varepsilon}(a_{x'}, x') \le -\frac{K-1}{K}(1-\varepsilon)h.$$

Since $(b_{x'}, x') \in \partial S \cap \mathrm{int}\,\Omega$, it is clear that

$$u(b_{x'}, x') - p_{y_\varepsilon}(b_{x'}, x') = 0.$$

Then by Jensen's inequality, we have

$$\begin{aligned}
&\int_{a_{x'}}^{b_{x'}} |D_{x_1}u(t,x') - D_{x_1}p_{y_\varepsilon}(t,x')|^q\ \mathrm{d}t \\
&\qquad \ge \frac{1}{(d_{x'})^{q-1}}\left(\int_{a_{x'}}^{b_{x'}} D_{x_1}u(t,x') - D_{x_1}p_{y_\varepsilon}(t,x')\ \mathrm{d}t\right)^q \\
&\qquad \ge \frac{1}{(d_{x'})^{q-1}}\left(\frac{K-1}{K}(1-\varepsilon)h\right)^q \\
&\qquad \ge \left(\frac{K-1}{K}(1-\varepsilon)\right)^q \frac{h^q}{r^{q-1}},
\end{aligned}$$

where $d_{x'} = b_{x'} - a_{x'} \le r$. Hence, there is

$$\begin{aligned}
\int_S |Du - Dp_{y_\varepsilon}|^q\ \mathrm{d}x &\ge \int_{\frac{1}{K}S} |Du - Dp_{y_\varepsilon}|^q\ \mathrm{d}x \\
&= \int_{P(\frac{1}{K}S)} \int_{a_{x'}}^{b_{x'}} |D_{x_1}u(t,x') - D_{x_1}p_{y_\varepsilon}(t,x')|^q\ \mathrm{d}t\ \mathrm{d}x'
\end{aligned}$$

$$
\begin{aligned}
&\geq \int_{P(\frac{1}{K}S)} \left(\frac{K-1}{K}(1-\varepsilon)\right)^q \frac{h^q}{r^{q-1}}\ \mathrm{d}x' \\
&= \left(\frac{K-1}{K}(1-\varepsilon)\right)^q \frac{h^q}{r^{q-1}} \left|P\left(\frac{1}{K}S\right)\right| \\
&= \frac{(K-1)^q(1-\varepsilon)^q}{K^{q+n-1}} \frac{h^q}{r^{q-1}} |P(S)| \\
&= \frac{(K-1)^q(1-\varepsilon)^q}{K^{q+n-1}} \frac{\varepsilon h^q}{r^q} (\varepsilon^{-1} r |P(S)|) \\
&\geq \frac{(K-1)^q(1-\varepsilon)^q}{K^{q+n-1}} \frac{\varepsilon h^q}{r^q} |S|. \qquad (2.9)
\end{aligned}
$$

Putting (2.9) and (2.8) into (2.4), we have

$$
\begin{aligned}
&\frac{1}{|S|}\int_S (F^1(x, Dp_{y_\varepsilon}) - F^1(x, Du))\ \mathrm{d}x \\
&\qquad = \frac{\delta}{|S|}\int_S (|Dp_{y_\varepsilon}|^q - |Du|^q)\ \mathrm{d}x \\
&\qquad\quad + \frac{1}{|S|}\int_S (G(x, Dp_{y_\varepsilon}) - G(x, Du))\ \mathrm{d}x. \\
&\qquad \leq \frac{\delta}{|S|}\int_S \left(2|Dp_{y_\varepsilon}|^q - \frac{1}{2^{q-1}}|Du - Dp_{y_\varepsilon}|^q\right)\ \mathrm{d}x \\
&\qquad\quad + \frac{1}{|S|}\int_S (G(x, Dp_{y_\varepsilon}) - G(x, Du))\ \mathrm{d}x. \\
&\qquad \leq 2\delta\left(\frac{\varepsilon h}{r}\right)^q - \frac{\delta(K-1)^q(1-\varepsilon)^q}{2^{q-1}K^{q+n-1}}\frac{\varepsilon h^q}{r^q} + Ch + C\frac{h^q}{r^{q-1}} \\
&\qquad = \left(2\varepsilon^{q-1} - \frac{(K-1)^q(1-\varepsilon)^q}{2^{q-1}K^{q+n-1}}\right)\delta\varepsilon\frac{h^q}{r^q} + Ch + C\frac{h^q}{r^{q-1}}.
\end{aligned}
$$

For fixed $\varepsilon > 0$ small sufficiently, we have $2\varepsilon^{q-1} - \frac{(K-1)^q(1-\varepsilon)^q}{2^{q-1}K^{q+n-1}} \leq -\widetilde{C}$ for some constant $\widetilde{C} > 0$ depending only on d and q, next choosing r_0 smaller such that $Cr \leq \frac{1}{2}\widetilde{C}\delta\varepsilon$. Then we know that there exist C_1, C_2 depending on d, q, δ, C_0 and M such that

$$
\frac{1}{|S|}\int_S \left(F^1(x, Dp_{y_\varepsilon}) - F^1(x, Du(x))\right)\ \mathrm{d}x \leq C_1 h - C_2\frac{h^q}{r^q},
$$

i.e. (2.2) holds. □

Remark 2.3. *From the proof of Lemma 2.1, we can see that C_1 is identical 0 if $G(x,\mathbf{p}) \equiv 0$, i.e., $F^1(x,\mathbf{p}) = \delta|\mathbf{p}|^q$. Hence,* (2.2) *becomes*

$$\frac{1}{|S|}\int_S (|y|^q - |Du|^q))\ \mathrm{d}x \le -C_2\frac{h^q}{r^q}. \tag{2.10}$$

2.3. **An interesting corollary.** From Remark 2.3, we can obtain a very interesting corollary of Lemma 2.1, which is also stated in [CL] for the $q = 2$ case.

Corollary 2.4. *Let $F^1(x,\mathbf{p}) = |\mathbf{p}|^q$ and $F^0(x,z) = f(x)z$ in* (1.2)*. Let u be a minimizer of* (1.2)*, then a non-trivial section of u can not be contained in the region where $f \le 0$. In particular, u is a ruled surface in the region where $f \le 0$.*

Before we give the proof, we recall a definition of extreme points of a convex set.

Definition 2.5 ([NB, Definition 9.2.1])**.** *A point p of a convex set K is an* extreme point *of K if p is not between any two distinct points of K. That is, if there does not exist $x, y \in K$ and $0 < t < 1$ such that $x \ne y$ and $p = tx + (1-t)y$.*

Proof of Corollary 2.4. We prove this corollary by contradiction. Suppose that there exists a non-trivial section of u is contained in $\{x \in X : f(x) \le 0\}$. Then there exists a point x' such that the contact set

$$\mathcal{T}_{x'} := \{x \in X : u(x) = u(x') + Du(x')\cdot(x - x')\}$$

contains at least one extreme point x_0 lying in the interior of $\{x \in X : f(x) \le 0\}$. Thus, for sufficiently small $r > 0$, we have

$$h := \sup_{B_r(x_0)} (u - l_{x_0}) > 0$$

and

$$\{x \in X : u(x) < l_{x_0}(x) + h\} \subset \{x \in X : f(x) \le 0\},$$

where

$$l_{x_0}(x) = u(x_0) + Du(x_0)\cdot(x - x_0).$$

Then, by Lemma 2.1 we choose an affine function $p_y(x)$ with the associated section $S = \{x :\ u(x) < p_y(x)\}$ and set

$$u_h := \max\{u, p_y\}.$$

It is clear that u_h is convex, then there is $L(u_h) \ge L(u)$ since u is a minimizer of (1.2). Note that u_h is different from u only on S. By (2.1), (2.10) in Remark 2.3 and

$$S \subset \{x \in X : u(x) < l_{x_0}(x) + h\} \subset \{x \in X : f(x) \le 0\},$$

we have

$$\begin{aligned} 0 \le L(u_h) - L(u) &= \int_S [(|y|^q + f(x)p_y(x)) - (|Du(x)|^q + f(x)u(x))]\,\mathrm{d}x \\ &= \int_S (|y|^q - |Du(x)|^q)\ \mathrm{d}x + \int_S (p_y(x) - u(x))f(x)\ \mathrm{d}x \end{aligned}$$

$$\leq -C_2 \frac{h^q}{r^q}|S|$$
$$< 0,$$

which makes a contradiction. Hence, any non-trivial section of u can not be contained in the region where $f \leq 0$. This implies that the contact sets of u have no extreme points inside $\{x \in X : f(x) \leq 0\}$, which gives us that u should be a ruled surface in $\{x \in X : f(x) \leq 0\}$. □

3. Optimal regularities

In this section, we will provide examples to demonstrate that the regularity in Theorem 1.1 is optimal for $q \geq 2$. When $q = 2$, there is an example constructed in [MRZ, Remark 5] to show that Theorem 1.1 is optimal.

As mentioned in the introduction, we do not have an explicit Euler-Lagrange equation for the minimizers of (1.1) with convexity constraints. In [Li], Lions has shown that the Euler-Lagrange equation for the minimizers of (1.1) with convexity constraints has the following form:

$$\sum_{i,j=1}^{n} \frac{\partial^2}{\partial x_i \partial x_j}\mu_{ij} = \frac{\partial F}{\partial z}(x, u(x), Du(x)) - \sum_{i=1}^{n} \frac{\partial}{\partial x_i}\left(\frac{\partial F}{\partial p_i}(x, u(x), Du(x))\right), \tag{3.1}$$

where $\mu = (\mu_{ij})$ is a matrix-valued Radon measure and (3.1) holds in the sense of distribution. See [Ca] for a different proof and some extensions. Since very little is known about the measure μ, the regularity of the minimizers of (1.1) with convexity constraints via (3.1) is still inaccessible. However, it was shown in [Ca] by using (3.1) that the minimizers of (1.1) with certain conditions on F must belong to the class of C^1 in the one-dimensional case.

Theorem 3.1 ([Ca, Theorem 3])**.** *Assume that $n = 1$ and suppose that $F(t, x, v)$ satisfies*

(1) F is of class C^1 over $(a, b) \times \mathbb{R} \times \mathbb{R}$,
(2) there exists $\beta > 0$, $\alpha \in L^{p'}((a, b))$ and $\gamma \in L^1((a, b))$ such that for all $(t, x, v) \in [a, b] \times \mathbb{R} \times \mathbb{R}$,

$$\left|\frac{\partial F}{\partial v}(t, x, v)\right| \leq \alpha(t) + \beta\left(1 + |v|^{p-1}\right),$$

$$\left|\frac{\partial F}{\partial x}(t, x, v)\right| \leq \gamma(t) + \beta\left(1 + |v|^{p}\right)$$

where we assume $p > 1$, and $\frac{1}{p} + \frac{1}{p'} = 1$,
(3) F is strictly convex with respect to v.

Then the minimizers of (1.1) with convexity constraints belongs to $C^1(a, b)$.

It can be observed that the assumptions outlined in **(H1)**-**(H3)** are partially inspired by Theorem 3.1 and have a certain naturalness in their formulation.

Let $\Omega = [-1, 1]$, $q > 2$, we consider the functional

$$L[u] := \int_{-1}^{1} \frac{1}{q}|u'(x)|^q + u(x)\ \mathrm{d}x \tag{3.2}$$

over the set

$$\{u : [-1, 1] \to \mathbb{R} \,|\, u \text{ is convex and } u(1) = u(-1) = 0\}.$$

By Theorem 3.1, we know that the minimizer of (3.2) with convexity constraints is already C^1. What's more, the minimizer of (3.2) is $C^{1,1/(q-1)}$ for $q > 2$ according to Theorem 1.1. Now, we show that when $q > 2$, the minimizer is at most $C^{1,1/(q-1)}$.

It is easy to see that the Euler-Lagrange equation of (3.2) without convexity constraint is

$$(|u'|^{q-2}u')' = 1. \tag{3.3}$$

Solving (3.3) with boundary conditions $u(-1) = u(1) = 0$ yields

$$u(x) = \frac{q-1}{q}\left(|x|^{1+\frac{1}{q-1}} - 1\right). \tag{3.4}$$

It is clear that u is a convex function on $[-1, 1]$, which implies that the minimizers of $L(u)$ with or without a convexity constraint coincide. Then we know that the regularity of $u(x)$ is at most $C^{1,1/(q-1)}$ when $q > 2$.

When $1 < q < 2$, we can observe that $u(x)$ in (3.4) is at least $C^{1,1}$ regularity, implying the inapplicability of our example in the $1 < q < 2$ case. However, the optimal regularity of u remains undetermined within this interval. Consequently, at the end of this section, we present two questions for future consideration. Firstly, what is the optimal regularity for the minimizer of (1.2) when $1 < q < 2$? Secondly, can the regularity established in Theorem 1.1 be extended to the boundary under certain boundary conditions?

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